

[Aim: 100 | 100 in Maths]

आरंभ

POLYNOMIALS

Lecture #6

ATK²:-

$$ax^2 + bx + c$$

$\swarrow$ $\searrow$
 α β

SOR

$$\alpha + \beta = -\frac{b}{a}$$

POR

$$\alpha \cdot \beta = \frac{c}{a}$$

Making of a Quad. Poly

We need $\Rightarrow S, P$

$$\text{Quad} \Rightarrow k[x^2 - Sx + P]$$

$$k \in \mathbb{R} - 0$$

#LP: The number of polynomials having α and β as zeroes is:

(A) 1

(B) 2

(C) 3

✓ (D) Infinitely many

$$S = \alpha + \beta \Rightarrow -2 + 5 \Rightarrow 3$$

$$P = \alpha \cdot \beta \Rightarrow -2 \times 5 \Rightarrow -10$$

$$\text{Quad} \Rightarrow k(x^2 - Sx + P)$$

$$\Rightarrow k(x^2 - 3x - 10)$$

$$k=1 \longrightarrow \checkmark$$

$$k=2 \quad 2x^2 - 6x - 20$$

$$k=3 \quad 3x^2 - 9x - 30$$

⋮

⋮

#LP: Construct a quadratic polynomial, the sum and product of whose zeroes are $-\frac{1}{4}$ and $\frac{1}{4}$ respectively. (CBSE 2017)

$$S = -\frac{1}{4}$$
$$P = \frac{1}{4}$$

$$\text{Quad} \Rightarrow k[x^2 - Sx + P]$$
$$\Rightarrow k\left[x^2 - \left(-\frac{1}{4}\right)x + \frac{1}{4}\right]$$

$$\Rightarrow k\left[x^2 + \frac{1}{4}x + \frac{1}{4}\right]$$

$$\Rightarrow k\left[\frac{4x^2 + x + 1}{4}\right]$$

$$\text{put } k \Rightarrow 4 \Rightarrow 4\left[\frac{4x^2 + x + 1}{4}\right]$$

$$\boxed{Q \Rightarrow 4x^2 + x + 1} \quad \checkmark$$

#LP:-The quadratic equation whose roots are

$$\frac{\sqrt{2} + \sqrt{3}}{\sqrt{2} - \sqrt{3}}$$

and

$$\frac{\sqrt{2} - \sqrt{3}}{\sqrt{2} + \sqrt{3}}$$

~~(A) $5x^2 - 2x + 5 = 0$~~

~~(B) $5x^2 + 2x + 5 = 0$~~

~~(C) $5x^2 + 2x - 5 = 0$~~

~~(D) $5x^2 - 2x - 5 = 0$~~

Final Q $\Rightarrow k(x^2 - Sx + P)$

$\Rightarrow k(x^2 - (-10)x + 1)$

$\Rightarrow k(x^2 + 10x + 1)$

$\Rightarrow S \Rightarrow 5(x^2 + 10x + 1)$

$\Rightarrow 5x^2 + 20x + 5$

$S \Rightarrow \alpha + \beta$

$$\Rightarrow \frac{\sqrt{2} + \sqrt{3}}{\sqrt{2} - \sqrt{3}} + \frac{\sqrt{2} - \sqrt{3}}{\sqrt{2} + \sqrt{3}}$$

$$\Rightarrow \frac{(\sqrt{2} + \sqrt{3})^2 + (\sqrt{2} - \sqrt{3})^2}{(\sqrt{2} - \sqrt{3})(\sqrt{2} + \sqrt{3})}$$

$$\Rightarrow \frac{(\sqrt{2})^2 + (\sqrt{3})^2 + 2\sqrt{2}\sqrt{3} + (\sqrt{2})^2 + (\sqrt{3})^2 - 2\sqrt{2}\sqrt{3}}{(\sqrt{2})^2 - (\sqrt{3})^2}$$

$$\Rightarrow \frac{2 + 3 + 2 + 3}{2 - 3}$$

$$\Rightarrow \frac{10}{-1} \Rightarrow \boxed{-10 = S}$$

$P = \alpha \cdot \beta$

$$= \frac{\sqrt{2} + \sqrt{3}}{\sqrt{2} - \sqrt{3}} \times \frac{\sqrt{2} - \sqrt{3}}{\sqrt{2} + \sqrt{3}}$$

$\boxed{P = 1} \checkmark$

#LP: A quadratic polynomial having zeroes $-\sqrt{\frac{5}{2}}$ and $\sqrt{\frac{5}{2}}$ is:

(A) $x^2 - 5\sqrt{2}x + 1$

(B) $8x^2 - 20$

(C) $15x^2 - 6$

(D) $x^2 - 5\sqrt{2}x - 1$

$$S = \alpha + \beta$$

$$= -\sqrt{\frac{5}{2}} + \sqrt{\frac{5}{2}}$$

$$S = 0$$

$$P = -\sqrt{\frac{5}{2}} \times \sqrt{\frac{5}{2}}$$

$$= -\left(\sqrt{\frac{5}{2}}\right)^2$$

$$P = -\frac{5}{2}$$

$$Q \Rightarrow k \left[x^2 - Sx + P \right]$$
$$\Rightarrow k \left[x^2 - 0x + \left(-\frac{5}{2} \right) \right]$$

$$= k \left[x^2 - \frac{5}{2} \right]$$

$$k = 8 \rightarrow 8 \left(x^2 - \frac{5}{2} \right)$$

$$= 8x^2 - \frac{5}{2} \times 8$$

$$= 8x^2 - 20 \quad \checkmark \text{ (B)}$$

#LP: If the zeroes of the polynomial $x^2 + px + q$ are double in value to the zeroes of the polynomial $2x^2 - 5x - 3$, then find the values of p and q.

#LP: Find a quadratic polynomial whose zeroes are the reciprocals of the zeroes of the polynomial $f(x) = ax^2 + bx + c$, $a \neq 0$, $c \neq 0$. [CBSE 2020]

$$\alpha, \beta$$

$$S = \frac{1}{\alpha} + \frac{1}{\beta}$$

$$S = \frac{(\alpha + \beta)}{(\alpha \beta)}$$

$$S = \frac{1}{\alpha \beta} \cdot \frac{\alpha + \beta}{1}$$

$$S = -\frac{b}{c}$$

$$P = \frac{1}{\alpha} \cdot \frac{1}{\beta}$$

$$P = \frac{1}{\alpha \beta}$$

$$P = \frac{1}{c/a}$$

$$P = \frac{a}{c}$$

$$ax^2 + bx + c$$

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha \cdot \beta = \frac{c}{a}$$

#LP: If α and β are zeroes of the polynomial $6y^2 - 7y + 2$, find a quadratic polynomial whose zeroes are $1/\alpha, 1/\beta$.

$$\rightarrow S = \frac{1}{\alpha} + \frac{1}{\beta} \Rightarrow \frac{(\alpha + \beta)}{\alpha\beta} \Rightarrow \frac{7/6}{2/6} \Rightarrow \boxed{\frac{7}{2} = S}$$

$$\rightarrow P = \frac{1}{\alpha} \cdot \frac{1}{\beta} \Rightarrow \frac{1}{\alpha\beta} \Rightarrow \frac{1}{2/6} \Rightarrow \frac{6}{2} \Rightarrow \boxed{3 = P}$$

$$Q = k(x^2 - Sx + P)$$

$$= k\left(x^2 - \frac{7}{2}x + 3\right)$$

$$k=2 \Rightarrow 2\left[x^2 - \frac{7}{2}x + 3\right] \Rightarrow \boxed{2x^2 - 7x + 6}$$

$$6y^2 - 7y + 2 \quad \swarrow \begin{matrix} \alpha \\ \beta \end{matrix}$$

$$\alpha + \beta = -\frac{(-7)}{6}$$

$$\boxed{\alpha + \beta = \frac{7}{6}} \quad \checkmark$$

$$\alpha \cdot \beta = \frac{2}{6}$$

$$= \frac{1}{3}$$

$$\therefore \boxed{\alpha \cdot \beta = \frac{2}{6}} \quad \checkmark$$

#LP: If α and β are zeroes of the quadratic equation $x^2 - 7x + 10$, find the quadratic whose zeroes are α^2 and β^2 .

$$S = \alpha^2 + \beta^2$$

$$S = (\alpha + \beta)^2 - 2\alpha\beta$$

$$S = (7)^2 - 2(10)$$

$$S = 49 - 20$$

$$S = 29$$

$$P = \alpha^2 \cdot \beta^2$$

$$= (\alpha \cdot \beta)^2$$

$$= (10)^2$$

$$P = 100$$

$\therefore Q = k(x^2 - 5x + 10)$
 $= k(x^2 - 29x + 100)$
 $k = 1$
 $\therefore Q = x^2 - 29x + 100$

$$x^2 - 7x + 10 \quad \swarrow \begin{matrix} \alpha \\ \beta \end{matrix}$$

$$\alpha + \beta = -(-7)$$

$$\alpha + \beta = 7$$

$$\alpha \cdot \beta = 10$$

$$\alpha \cdot \beta = 10$$

#LP: Find the zeroes of the polynomial $r(x) = 4x^2 + 3x - 1$. Hence, write a polynomial whose zeroes are reciprocal of the zeroes of polynomial $r(x)$. [CBSE 2025]

$$4x^2 + 3x - 1 = 0$$

$$4x^2 + 4x - x - 1 = 0$$

$$4x(x+1) - 1(x+1) = 0$$

$$(4x-1)(x+1) = 0$$

$$4x-1=0 \quad | \quad x+1=0$$

$$x = \frac{1}{4}$$

$$x = -1$$

Reciprocal $\rightarrow$ $\frac{1}{\alpha}$, $\frac{1}{\beta}$

$$S = 4 + (-1) \Rightarrow 3 = S$$

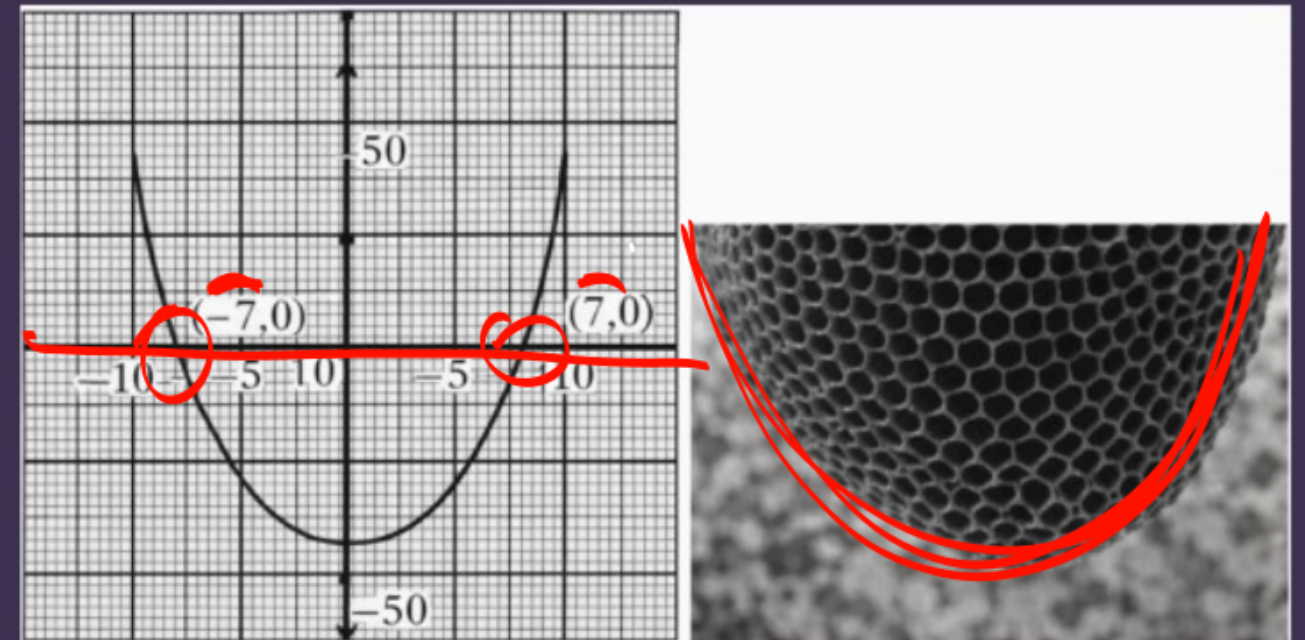
$$P = 4 \times (-1) \Rightarrow -4 = P$$

$$Q = k(x^2 - Sx + P)$$
$$= k(x^2 - 3x - 4)$$

#LP: While playing in a garden, Samaira saw a honeycomb and asked her mother what is that her mother replied that it's a honeycomb made by honey bees to store honey. Also, she told her that the shape of the honeycomb formed is a mathematical structure. The mathematical representation of the honeycomb is shown in the graph. Based on the above information, answer the following question: (CBSE 2023)

1. How many zeroes are there for polynomial represented by the graph given?

↪ 2 zeroes.



2. Write the zeroes of the Polynomial.

$$\text{Value of zeroes} \rightarrow \{-7, 7\}$$

3. If the zeroes of a polynomial $P(x) = x^2 + (a+1)x + b$ are $\checkmark 2$ and $\checkmark -3$ then determine the values of a and b

$$P(2) = 0$$
$$2^2 + (a+1)2 + b = 0$$

$$4 + 2a + 2 + b = 0$$

$$2a + b + 6 = 0$$

$$2a + b = -6$$

$$P(3) = 0$$

$$(3)^2 + (a+1)3 + b = 0$$

$$9 + 3a + 3 + b = 0$$

$$3a + b + 12 = 0$$

$$3a + b = -12$$

Hw → Module.

THANK YOU COODIESS !!

